AI for Scientific Writing — Session 2 handout

Topic card · search string · six questions · literature matrix · prompts — Semana Tec, Tec de Monterrey, Oct 2026

lrodriguez15.github.io/ai-scientific-writing.html

1. Topic card (Activity 4 — validate it with your professor)

Field	Your answer
Topic (specific: not “renewable energy” but “soiling losses in rooftop panels in Querétaro”)	
Question (what you want to answer)	
Objective (one sentence, a checkable verb: compare, measure, review…)	
Type (review · research · brief summary)	
Audience (who will read it)	
Scope (what is in, what is out)	

The same five facts start your conversation with **RigorIA** in Task 4.

2. Search string (Activity 5)

Main concept + synonyms (OR)	
Second concept + synonyms (OR)	
Third concept (AND)	
Final string	
Filters (years, type, language)	
Where you ran it	☐ Biblioteca Digital Tec ☐ Google Scholar ☐ IEEE Xplore ☐ Scopus ☐ SciELO / Redalyc

Example: `("solar panel" OR photovoltaic) AND (cleaning OR soiling) AND robot*` — quotes = exact phrase, OR = synonyms, AND = must have, * = any ending. AI can suggest synonyms and keywords; it is **not** the source of the reference.

3. Is this source reliable? — six questions

1. **Does it exist?** The DOI opens at doi.org; same title, authors and year.
2. **Who wrote it?** Authors and institution you can find.
3. **Where?** A peer-reviewed journal or conference — not a predatory journal or a content farm.
4. **When?** Recent enough for the topic.
5. **Evidence?** Data and methods, not only opinions.
6. **Does it say that?** You read the abstract *and* the results.

Peer-reviewed ≠ grey literature (theses, reports, news, blogs): grey literature gives context, not your main claims. **An AI answer is not a source** — the source is what it points to, after you open it.

4. Literature matrix (at least 5 sources: ≥ 3 peer-reviewed, ≥ 1 in English)

#	Reference (author, year, journal, DOI)	Main idea	Method	Key result	For my article	Verified
1					supports / contrasts / context	DOI opened ☐
2						☐
3						☐
4						☐
5						☐

Group by theme, not by source ("two studies measured X [1], [3]; one disagrees [2]"). **Summarise in your words.**

5. Prompts for today and Tasks 3–4

Spelling and grammar only: *Correct spelling and grammar only. Do not change the meaning, do not add content, keep technical terms. List every change you made. Text: "..."*

Translation: *Translate into Spanish for an engineering audience. Keep technical terms; list the terms you were unsure about.* (Check technical terms: in mechanics, "stress" is *esfuerzo*, not "estrés".)

Integrating sources (grounded): *Below are the abstracts of my 5 sources, numbered [1]–[5]. Group them by approach. List where they agree and disagree, citing the source number for every point. Use ONLY these abstracts. Do not add sources. Say if something is not in them.* — then open every source it cites and check.

Task 4 — RigorIA (MiTec → TECgpt → Agent Studio → RigorIA): *I am writing a [review / research / brief] article on ____. The objective is ____. Audience: ____. Suggest a detailed structure with the usual sections (Introduction, Methods, Results, Discussion, Conclusions) and suggested subsections.* Then section by section: *Write a 350-word introduction that includes the context of the problem, relevant background and the objective of the article.* Revise every section yourself; log every prompt.

6. Principles of technical writing — checklist

☐ one idea per sentence ☐ numbers with units (and n, uncertainty) ☐ no "very / a lot / revolutionary" ☐ every claim cited or measured ☐ terms and acronyms defined on first use ☐ consistent tense, units and citation style ☐ limitations and AI use declared